Kevin Stachelek

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Bioinformatics scientist with a PhD in Cancer Biology and Genomics who runs and maintains production analysis pipelines across single-cell RNA-seq, spatial transcriptomics, epigenomics, and multi-omics data. At a university genomics core I process sequencing data as it comes off the instruments, deliver quality-controlled results to wet-lab collaborators, and version everything through GitHub. I work in Python, R, git, and Linux with strong applied-statistics and data-science skills, and I develop with AI-assisted tools including Claude Code.

Core Competencies

- **Analysis pipelines**: implement, test, and maintain reproducible high-throughput pipelines (Snakemake, Nextflow, Docker/Apptainer, HPC/Slurm, Illumina DRAGEN); run production datasets and provide QC from instrument output onward.
- **Single-cell & multi-omics**: scRNA-seq (10x, Parse), spatial transcriptomics (Xenium, Visium HD), bulk ATAC-seq and BS-seq epigenomics, and integrated multi-omics (RNA + proteomics + CUT&RUN); clustering, Harmony/scVI integration, trajectory inference, cell2location, Moran's I.
- **Programming**: high competency in Python (Scanpy, anndata, spatialdata, squidpy) and R (Seurat, Bioconductor, Shiny); git/GitHub, Linux; Rust/PyO3 for accelerating compute hot loops.
- **AI-assisted development**: build and adapt pipelines with Claude Code and agentic tooling; delivered a Snakemake bulk RNA-seq workflow this way on a local DRAGEN server.
- **Imaging & translational genomics**: imaging-based spatial data with cell segmentation (Cellpose, StarDist, Baysor); cancer genomics and rare-disease/neurodegeneration work (ALS, cerebellar ataxia) with wet-lab and clinical collaborators.

Professional Experience

- **UCI Genomics Research and Technology Hub** — **Irvine, CA**
  Bioinformatician — *2025–Present*
  - Implement, test, and maintain production pipelines across 10x/Parse scRNA-seq, Xenium spatial, bulk RNA-seq, ATAC-seq, and BS-seq; process data as it comes off the sequencers (demultiplexing on DRAGEN and containerized HPC3) and provide quality assurance.
  - Analyze large-scale single-cell, spatial, and multi-omics datasets for many concurrent projects; deliver differential expression and interactive Seurat/Shiny portals to wet-lab scientists.
  - Build and adapt pipelines with AI-assisted tooling (Claude Code), and maintain all deliverables as public, versioned GitHub repositories.
  - Support rare-disease and neurodegeneration genomics (ALS de novo assembly, cerebellar ataxia panel) and optimize HPC memory for 100 GB+ imaging-based spatial datasets.
- **USC / Children's Hospital Los Angeles (Cobrinik Lab)** — **Los Angeles, CA**
  Graduate Student Researcher — *2018–2025*
  - Developed a computational pipeline to call somatic variants from cancer genomes with matched controls; integrated public sequencing data to identify recurrent drivers (first-author paper, 2023).
  - Applied single-cell methods (Seurat, Monocle, Scanpy, SCENIC) to define developmental trajectories and model tumor initiation.
  - Validated candidate driver mutations with luciferase reporters and CRISPR base editing.

- **Children's Hospital Los Angeles** **Los Angeles, CA**
  Research Technician *2013–2018*
  - Provided bioinformatic cfDNA copy-number analyses for cancer liquid-biopsy studies across multiple publications, working directly with clinical and wet-lab teams.

Education

- **University of Southern California** **Los Angeles, CA**
  PhD, Cancer Biology and Genomics *2018–2025*
  Single-cell and exome genomics of cancer progression; somatic copy-number alterations and driver-variant discovery.
- **University of Southern California** **Los Angeles, CA**
  MS, Translational Biotechnology *2015–2018*
- **University of Southern California** **Los Angeles, CA**
  BA, Biological Sciences *2009–2013*

Selected Publications

- Stachelek K, et al. (2023). Non-synonymous, synonymous, and non-coding nucleotide variants contribute to recurrently altered biological processes during retinoblastoma progression. *Genes, Chromosomes and Cancer* 62(5):275–289.
- Li HT, ...Stachelek K, ...Berry JL (2022). Characterizing DNA methylation signatures of retinoblastoma using the aqueous humor liquid biopsy. *Nature Communications* 13:5523.
- Cambier L, Stachelek K, et al. (2021). Extracellular vesicle-associated repetitive element DNAs as candidate osteosarcoma biomarkers. *Scientific Reports* 11:94.
- Singh HP, ...Stachelek K, ...Cobrinik D (2018). Developmental stage-specific proliferation and retinoblastoma genesis in RB-deficient human cone precursors. *PNAS* 115(40):E9391–E9400.

Honors and Awards

- Poster Presenter – American Association for Cancer Research (AACR) (2024).
- Poster Presenter – Keystone Symposia, Single Cell Biology (2019).
- National Merit Scholarship (2009–2013).

References

- Available upon request.